

MEWP telematics tracker — design verification report

Board **telematics-tracker**, 100 × 68 mm, 4-layer (JLC04161H-7628, ENIG) · schematic and board at tag **v1.0.2-verified-2026-09-30** (branch `relief6-stretch`) · review date 2026-09-30 · prepared by Claude Code for Immanuel David · manufacturing package: `out/telematics-tracker-mfg-2026-09-30-<commit>.zip`

Verdict. The design as packaged on 2026-09-29 (v1.0.1) would **not** have worked on arrival: five faults would have killed the board or a whole function at the first power-up (input TVS wired forward, 3.3 V rail shorted through two clamp diodes, both digital inputs blind, modem UART permanently disabled, buck current limit disabled), and a reversed supply would have destroyed it. All twelve faults found are **fixed in this revision**, re-verified, and every machine check is clean: schematic ↔ board parity 0 mismatches (212 parts, 727 pad connections), ERC 0 errors, DRC 0 errors / 0 unconnected, pin checklist pass, BOM 76/76 lines available at JLCPCB.

Ready to order for the prototype batch. Before a production release, seven bench measurements (section 6) and the certification tests in section 7 remain — those cannot be closed by review.

Legend: **OK** verified correct **FIXED** was NOT OK, corrected and re-verified in this revision **VERIFY** correct by design, must be measured or tested **NOT OK** open, needs action **N/A** not applicable.

1. Summary by area

Area	Result	Evidence
All parts placed and connected	OK	212 schematic parts = 212 board footprints (+5 mounting holes); 727 pad-to-net connections identical on both sides; 0 unconnected items; every handoff \$5 MCU pin on its net (tools/checkpins.py); no dangling nets other than the 7 known single-pin nets (spares, U6 STAT/PG, sense labels).
Schematic correct against datasheets	FIXED	Every IC pin checked against its datasheet; 12 faults found (section 3), all corrected in the generator (tools/sheets.py) and on the board, regenerated and re-checked.
Board rules (DRC)	OK	0 errors, 0 unconnected under the LV-127 rule set plus the HV creepage rules (.kicad_dru). 11 warnings, all harmless (8 silkscreen marks on the DNP eSIM pads, 1 silk text touching D2's outline, 2 KiCad "dangling" flags on connected T-junctions).
Current capacity / thermal	FIXED	Charger nets were 0.15 mm (≈ 0.18 A) for 0.69 A: now 0.6 / 0.4 / 0.3 mm. 5V0, SYS, 3V3, VIN, VBAT_MODEM trunks 0.5–2.0 mm. Buck exposed pad: 9 thermal vias to its inner-layer island.
High-voltage clearance and creepage (100 V)	OK	HV nets in their own class (0.6 mm), 1.5 mm HV-to-signal inside the HV zone, 1.0 mm to the board edge, conformal coating required (F-20) — all enforced by DRC rules, 0 violations.
Decoupling, crystals, RF	FIXED	C61 (charger input cap) moved from 49 mm to 11 mm from U6; 28 GND stitching vias added along both 50 Ω antenna lines; crystal load caps and placement OK.
BOM (availability, footprint, ratings)	OK	tools/bom_audit.py: 76 lines, every LCSC number resolved, in stock, package matches footprint, 0 fixes needed (docs/bom-audit.md). X5R caps replaced by X7R (rule 1).
Debug, flash and bring-up path	OK	SWD, ROM-bootloader UART, modem USB, modem UART monitor and every rail on test points (section 5).

Enclosure IP65	VERIFY	Housing (Blender → STEP/STL, two variants) checked against the assembled-board STEP; gland, gasket groove, antenna lid. Ingress rating is proven only by an IEC 60529 test.
Industry / regulatory requirements	NOT OK	Design provisions are in place; no EMC, radio, safety, ingress or battery certification test has been run yet (section 7). Required before selling.

2. What was checked and how

Check	Method	Result
Schematic ↔ board parity	Netlist export vs. board pads: reference, footprint, value, DNP flag and the net of every pad	0 mismatches
MCU pin map	Handoff §5 table vs. netlist (tools/checkpins.py), plus F-10 default-OFF structure	all pins on their nets, pass
ERC	kicad-cli, all severities	0 errors, 23 known warnings
DRC	kicad-cli, all severities, LV-127 rules + HV .kicad_dru	0 errors, 0 unconnected, 11 warnings
Datasheet review of every block	Pin-by-pin against the manufacturer PDFs (Artery, Quectel HW Design V1.2, EG Micro, TI, Nexperia, SIT, Everlight, Analog Power, GigaDevice, QST, ST, Hirose/JXTCONN); KiCad library pin names read from the .kicad_sym files; independent second pass on every finding	12 faults → fixed
Current capacity	IPC-2221 for track width / layer / copper weight vs. worst-case current on every power net	3 nets widened
Copper path resistance	Graph of tracks/vias/pours, shortest resistive path per rail	no bottlenecks left
Clearance / creepage	DRC rule set; measured minimum distances HV ↔ LV, HV ↔ edge	OK
Decoupling distance	Cap-to-pin distance for every decoupling cap	1 moved
RF	CPWG geometry on the declared stackup (0.2104 mm dielectric), stitching, keep-outs	stitching added
BOM	LCSC + JLCPCB APIs: stock, package, key attributes per line	76/76 clean
Mechanical	Board STEP inside the housing model, part heights, gland/locknut, lid antennas	OK by model

3. Faults found and fixed (all would have failed on the bench)

#	Item	What would have happened	Root cause	Fix (schematic / board)
1	D2 SMDJ100A input TVS	Conducts forward at the first power-up: input shorted through the TVS, F1 opens. Dead on arrival.	Drawn with KiCad's bidirectional D_TVS symbol; pin 2 on the rail, but on the SMC footprint pad 1 is the cathode (band).	Symbol D_Zener, cathode (pin 1) to VIN_P, anode to GND. Board: D2 turned 180°.

2	D10 / D11 BAV99 clamps on VIN_SENSE / IGN_SENSE	Both diodes forward between 3V3 and GND: 3.3 V rail held at ≈ 1.2 V, MCU never starts. Dead on arrival.	KiCad's Diode:BAV99 symbol carries hidden pin names K/A/K that contradict its own drawing and Nexperia's datasheet (pin 1 = A1, pin 2 = K2, pin 3 = common).	Pin 1 (anode) $\rightarrow$ GND, pin 2 (cathode) $\rightarrow$ 3V3, pin 3 signal. Board: both turned 180°; D10 moved 0.3 mm, NRST via relocated, VIN_SENSE re-fed.
3	D5 / D6 BAV99 across the DI opto LEDs	Bypasses the opto LED on every positive input: DI1 and DI2 never read. Function dead.	Same pin-name trap, wired for the mirror of the real pinout.	BAV70 (common cathode) with the same three connections — pin-compatible, no copper change.
4	U8 TXB0104 output enable	Modem UART, RI and DTR permanently disabled: no modem communication at all.	OE divider 47 k up / 10 k down = 0.32 V; TXB0104 needs $\geq 0.65 \times V_{CCA} = 1.17$ V.	R63 10 k up, R62 100 k down $\rightarrow$ 1.64 V (Quectel uses 10 k / 120 k).
5	R82 buck current-sense resistor	No over-current or short-circuit protection on the 5 V converter; a shorted 5V0 destroys U5/L1.	Fitted 0 R (F-12: "no formula in the datasheet"). The EG11752 compares (IS – VS) with 0.2 V, so $R_{IS} = 0.2 \text{ V} / I_{\text{peak}}$.	0.1 R (YAGEO RL0805FR-7W0R1L) $\rightarrow$ 2 A peak limit, normal peak ≈ 1.2 A, L1 saturation 2.7 A. Bench BV-1.
6	Reverse-polarity protection	A reversed supply flows GND $\rightarrow$ Q1/Q2 body diode $\rightarrow$ D7/D8 $\rightarrow$ VIN around the D1 blocking diode with nothing limiting it: Q1/Q2, D7/D8 destroyed, then F1.	Flyback diodes returned to raw VIN instead of the protected node.	D7/D8 cathodes to VIN_P (behind D1), exported as a hierarchical net. Board: VIN trunk re-laid around the diode pads, new 0.5 mm VIN_P run to D1. Bench BV-6.
7	Q1 / Q2 gate drive	DO1/DO2 MOSFETs only partly enhanced: hot, may not switch a 0.5 A load.	10 k gate pulldown loaded the 2.2 k / 100 R driver: 4.07 V at the gate vs. the AM2390N's 4.5 V specification.	R25/R26 100 k $\rightarrow$ 4.89 V, gate still discharges in ≈ 25 μ s.
8	DI1 / DI2 pull-up	Inputs read "off" while driven at the low end of the range or when hot.	47 k pull-up needs 70 μ A from the opto; at 0.22–0.26 mA LED current the EL357N(D) delivers about that ($\approx 1\times$ margin).	R21/R22 220 k (15 μ A needed, $\approx 7\times$ margin). Bench BV-7.
9	SIM card detect	USIM_DET floats: hot-plug detection unusable.	The EC200U has no internal pull-up on USIM_DET (Quectel Fig 18 shows 51 k to VDD_EXT).	R91 51 k VDD_EXT $\leftrightarrow$ USIM_DET added (bottom side, both pads on existing copper). Firmware FW-17.
10	SIM ESD array	SIM signal edges loaded by 80–130 pF per line; risk of SIM errors.	SMF05C chosen; Quectel asks for ≤ 15 pF.	PESD5V0L5UY (≤ 19 pF, same SOT-363 pinout). 19 vs 15 pF accepted: nearest in-stock part, SIM clock ≤ 5 MHz.
11	Ceramic capacitor dielectric	X5R 1 μ F / 4.7 μ F lose most of their capacitance with bias and temperature (rule 1 of the handoff).	Library defaults.	1 μ F $\rightarrow$ YAGEO CC0603KRX7R8BB105 (X7R 25 V), 4.7 μ F $\rightarrow$ CC0805KKX7R8BB475 (X7R 25 V).

12	Charger power copper	VBAT_BT ran 13 mm on the inner layer at 0.15 mm (≈ 0.18 A) for the 0.69 A charge current: overheating trace; SW_CHG and PMID also 0.15 mm.	Router class assignment.	VBAT_BT 0.6 mm on F.Cu U6 → J2, SW_CHG 0.4 mm, PMID 0.3/0.5 mm.
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Also corrected on the board: X2, R70, C48–C51 marked DNP so pick-and-place skips them; C61 (charger VBUS cap) swapped with C80 to sit 11 mm from U6; 28 GND stitching vias along both antenna lines; 4 more thermal vias under U5. Documented rather than changed (section 8): the 10.5 V input floor, ignition wake by ADC polling, the ground-referenced (not isolated) digital inputs, and the SYS 3.5–4.26 V range.

4. Verified OK, block by block

Block	Checked	Result
Input front end	F1 1 A 250 VAC / 125 VDC fuse (100 A DC interrupt) → D1 S3M reverse block → D2 SMDJ100A 3 kW TVS (now correctly oriented) → R80 1 R 2512 surge resistor → C70/C71 2.2 μ F 100 V + C72 100 nF. Clamp margin at 3.7 A: 34.9 %. TP23 on VIN.	OK
5 V buck U5 EG11752	Every pin per datasheet: EN 100 k from VCC (fig 6-2), VB–VS 100 nF bootstrap, FB 4.3 k / 1.5 k → 5.03 V, L1 150 μ H (Isat 2.7 A), D3 SS3200 free-wheel, IS 0.1 R (2 A), exposed pad = VIN_B on its own inner island with 9 vias, 0.6 mm clearance inside the BUCK_HV area, output 4 × 10 μ F + 100 nF, TP19.	OK
Charger / power path U6 BQ25606	Pin 2 NC per datasheet, ICHG 0.69 A, ILIM 0.89 A, VSET floating = 4.208 V, SYSMIN 3.5 V, BTST 47 nF, L3 2.2 μ H, PMID / SYS / VBUS capacitors, TS JEITA divider R88/R89 with the pack NTC (charge inhibited outside 0–45 °C), STAT/PG left open, J2 battery connector (B+, NTC, B–), TP13 on SYS.	OK
3.3 V LDO U9 ME6211	Pinout, input/output capacitors, 500 mA class vs. the 3V3 load, TP6.	OK
MCU U2 AT32F403ACGT7	47/47 handoff \$5 pins on their nets; BOOT0 10 k + JP1, BOOT1 low; NRST 100 nF + TP3; 8 MHz crystal + 2 × 18 pF and 32.768 kHz + 2 × 6.8 pF; VDDA ferrite + 1 μ F/100 nF; 100 nF at every VDD pin; SWD on TP1/TP2; USART2 (ROM bootloader) on TP4/TP5; CAN1 remap PB8/PB9; PC13 LED within its 3 mA limit; JTAG pins reused as GPIO (FW-1).	OK
Modem U1 Quectel EC200U-CN	All 112 pads: VBAT (4 × 47 μ F X7R + 1 μ F + 100 nF + TVS within 5 mm), all GND pads tied, PWRKEY and RESET_N through NPN inverters, STATUS through NPN (inverted, FW-5), NETLIGHT → NPN → LED, VDD_EXT 1.8 V → U8, USB_DP/DM/VBUS → USBLC6 → TP15–17, USIM (33 R series, 100 nF, PESD, 51 k detect pull-up), antennas 50 Ω CPWG with π pads to U.FL (AF1 LTE, AF2 GNSS), GNSS active-antenna bias L4/R90/C83/C85, MODEM_PWR_EN P-FET power cut (FW-6).	OK
Level shifter U8 TXB0104	A-side 1.8 V (VDD_EXT), B-side 3.3 V, OE 1.64 V (fixed), TX/RX/RI/DTR channel mapping to PA9–PA12, unused channels tied.	FIXED
CAN U4 SIT1051	VCC 5V0, VIO 3V3, STB → PA15 (FW-18), TXD/RXD to the remapped CAN1, common-mode choke (1–4, 2–3), PESD1CAN TVS at the connector, split termination 2 × 60 R + 4.7 nF behind JP2 (open by default), TP25/TP26.	OK
Digital inputs	36 k (3 × 12 k 1206) → EL357N(D) opto + BAV70 reverse diode, 220 k pull-up + 100 nF to PB12/PB13; valid 10.5–100 V; ground-referenced.	FIXED
Digital outputs	F-10 two-stage NPN driver from 5V0 (provably off with the MCU unpowered), AM2390N 100 V MOSFET, 100 R gate, 100 k pulldown, SS3200 flyback to VIN_P; rated 0.5 A at 12/24 V.	FIXED

Sense dividers	VIN and IGN $3 \times 100 \text{ k} : 9.1 \text{ k} (\div 34; 100 \text{ V} \rightarrow 2.9 \text{ V})$ with BAV99 clamps to 3V3/GND (fixed) and 100 nF; VBAT 1 M : 1 M from SYS.	FIXED
Storage U7 GD25Q64E (I-grade)	SPI1 pins, WP#/HOLD# high, 100 nF + 1 μ F, CS pull-up.	OK
IMU U3 QMI8658B	I2C address 0x6A (SA0 high), CS high = I2C mode, INT1 $\rightarrow$ PB5, decoupling, 2.2 k I2C pull-ups.	OK
Connectors	J1 Micro-Fit 12-pin pin map (VIN, GND, IGN, CANH, CANL, GND, DI1, DI2, DO1, DO2, 2 spares) matches the installation sheet; J2 battery; X1 SIM holder pin map incl. CD switch; U.FL $\times 2$.	OK
Layout	Solid GND on L2, 3V3/SYS pours on L3, HV zone $x < 44 \text{ mm}$ with the 1.5 mm rule, LV-127 rules at JLCPCB standard capability (0.127 mm clearance, 0.15 mm track, 0.2 mm drill), staircase tracks straightened, 5 GND-tied M3 holes, silk references on F.Fab for passives (assembly drawing).	OK
Enclosure	IP65 housing model (two variants: internal lid antennas / external SMA), M16 gland, cavity clearances measured against the board STEP (12.1 mm over J1, 2.4 mm under bottom parts, gland locknut clear), STL meshes manifold.	VERIFY

5. Development: debug, flash and first power-up

Interface	Test points / parts	Use
SWD (primary programming and debug)	TP1 SWDIO, TP2 SWCLK, TP3 NRST, TP6 3V3 (target reference), TP7 GND	Artery AT-Link, J-Link or a DAPLink probe with Artery's ICP tool / OpenOCD (AT32 support). Full flash, breakpoints, RTT.
ROM bootloader (serial ISP, no debugger)	JP1 solder jumper = BOOT0 high; TP4 DBG_TX (MCU PA2, USART2 TX), TP5 DBG_RX (PA3), 3.3 V logic, TP7 GND	Artery ISP programmer over any 3.3 V USB-UART. Open JP1 again after flashing. The same pads are the application's console port.
Modem USB	TP15 USB_DP, TP16 USB_DM, TP17 USB_VBUS (5 V in), TP18 GND	Quectel QFlash firmware update, QNavigator AT console, USB modem/ECM test, DFOTA setup.
Modem UART monitor	TP27 MODEM_TX, TP28 MODEM_RX (3.3 V side of U8)	Sniff the MCU $\leftrightarrow$ modem AT traffic with a logic analyser.
Power rails	TP23 VIN, TP19 5V0 (5.03 V), TP13 SYS (3.5–4.26 V), TP24 VBAT_MODEM, TP6 3V3 (3.30 V), modem VDD_EXT 1.8 V at C44	First power-up checklist; current-limited bench supply 24 V / 0.5 A, battery disconnected.
CAN	TP25 CANH, TP26 CANL; JP2 closes the split termination	Bus tests with a CAN analyser; close JP2 only on a bench bus without terminators.
Indicators	SYS_LED (PC13 heartbeat), NET_STATUS_LED (modem NETLIGHT)	Boot and registration at a glance.

First power-up order (details in docs/bringup-test-plan.md): 5V0 $\rightarrow$ SYS $\rightarrow$ 3V3 with the modem held off (PB10 high), then release the modem, expect STATUS low (FW-5) and VDD_EXT 1.8 V $\approx$ 4–5 s after PWRKEY, then AT on TP27/TP28. Firmware constraints that hardware imposes are listed as FW-1 ... FW-22 in docs/firmware-notes.md (JTAG disable, CAN remap, CAN standby pin, modem power sequence, OE behaviour, no internal pull-ups on the modem UART pins, ADC-pollled ignition wake, IWDG in sleep, battery floor, LTE-only scan, SIM detect polarity).

6. Bench measurements required before production

#	Measurement	Why / pass criterion
BV-1	Buck current limit (short 5V0 through 0.1 R at 100 V in)	Converter folds back without damage; trip near 2 A peak.
BV-2	Buck minimum on-time at 100 V in, 0.7–1 A out, and light load	Stable 5.03 V, no pulse skipping instability ($T_{on} \approx 455$ ns).
BV-3	EN pin voltage across the VCC range	Within the 7 V absolute maximum.
BV-4	U5 qualification: 3 units, 1000 h at 100 V / 85 °C, thermal cycling $-40 \leftrightarrow +85$ °C	No drift, no failure (U5 approval condition).
BV-5	Input dropout: VIN at which 5V0 leaves regulation at 1 A	Expected 10.5–12 V; sets the firmware "on battery" threshold (FW-13).
BV-6	Reverse polarity –24 V with the DO loads connected	No current, no damage (fix 6).
BV-7	DI threshold at 10.5 V, –20 °C and +70 °C	PB12/PB13 read low with margin (fix 8).
—	Thermal soak at 60 °C ambient with modem transmitting and the battery charging	Modem ≤ 70 °C case, buck and charger within ratings (handoff thermal rule).
—	Antennas: GNSS fix and LTE registration inside the closed housing; GNSS $\leftrightarrow$ LTE isolation > 40 dB	Cold fix < 60 s outdoors; RSRP within 3 dB of a reference antenna.
—	Battery: charge/discharge cycle, JEITA inhibit outside 0–45 °C, ride-through on VIN loss	Per handoff §4.

7. Industry and regulatory requirements

Requirement	Status	Design provision / what is still needed
RoHS (2011/65/EU) and lead-free assembly	VERIFY	JLCPCB assembles lead-free; every LCSC line is a RoHS-marked part. Collect the assembler's and part suppliers' declarations for the technical file.
EMC — emissions and immunity (CE: EN IEC 61000-6-2 / 61000-6-4; machinery: ISO 13766 for construction machinery, EN 280 for MEWPs)	NOT OK	Provisions: input TVS and surge resistor, CAN common-mode choke + TVS, ESD arrays on SIM/USB/CAN, solid ground plane, filtered sense inputs, conformal coating. Not yet tested — book a pre-compliance scan (radiated/conducted emissions, ESD, EFT, surge, RF immunity).
Automotive/mobile-machine supply transients (ISO 7637-2 pulses, ISO 16750-2 load dump, reverse battery)	VERIFY	100 V standoff, 3 kW TVS, 1 R surge resistor, reverse blocking diode + fixed flyback return. Pulse test on the bench (BV-6 plus ISO 7637 pulses 1, 2a, 3a/3b).
Radio approval — LTE module (India WPC ETA; elsewhere CE-RED / FCC via the module's grants)	VERIFY	EC200U-CN covers India's LTE bands (B1/B3/B5/B8/B40/B41). The module is pre-certified by Quectel; confirm with the Quectel India distributor that the EC200U-CN variant's WPC ETA is on file (or file the self-declaration ETA), and use the module's certified antenna gain limits.
India TEC MTCTE (telecom equipment)	VERIFY	Cellular IoT/tracking devices may fall under MTCTE. Confirm the applicable category with a TEC-designated lab before sale.

AIS-140 (vehicle tracking, India)	N/A	Applies to transport-vehicle tracking devices, not to MEWP fleet telematics. If a customer demands it: AIS-140 adds embedded dual-profile SIM, 4 h backup, ARAI/ICAT type approval — not covered by this design.
Electrical safety (IEC 62368-1): 100 V DC input is above the 60 V ES1 limit	VERIFY	The HV zone is enclosed (IP65 housing), coated (F-20) and separated by the DRC creepage rules; no user-touchable HV parts. Formal classification by a safety lab if the product is certified.
Ingress protection IP65 (IEC 60529)	VERIFY	Housing with gasket groove, M16 cable gland, lid antennas under the cover. Needs the IP6X dust and IPX5 water-jet test on a finished sample.
Environmental: temperature, vibration, shock (ISO 16750-3/-4, IEC 60068-2-6/-27/-64)	NOT OK	Product rating $-20 \dots +60 \text{ }^{\circ}\text{C}$; five M3 mounts; only J1/J2 through-hole. No vibration or thermal-cycling test yet.
Battery safety and transport (IEC 62133-2 cell, UN 38.3, India BIS CRS under IS 16046 for Li-ion cells/packs)	NOT OK	1S 103450 pack with NTC and JEITA charging is the right architecture; the cell/pack itself must come with IEC 62133-2, UN 38.3 and (for India) BIS registration from the pack vendor.
Conformal coating	NOT OK	Mandatory for the 100 V creepage (F-20). Specify it in the assembly order (acrylic or silicone, mask the connectors, U.FL, test points and the SIM holder).
Marking and documentation	NOT OK	Rating label (12–100 V DC, IP65, model, serial), installation sheet (docs/installation-sheet.md, corrected), declaration of conformity once the tests above pass.

8. Open items (NOT OK or accepted deviations)

Item	Status	Detail / owner
Guaranteed input floor	VERIFY	Specified 10.5 V; after D1 and R80 the buck sees $\approx 9.2 \text{ V}$ at full load, under its 10 V minimum. Documented as 12 V guaranteed / 10.5 V bench figure (BV-5, FW-13, installation sheet). Fix option if the bench fails: $R80 \ 1 \ R \rightarrow 0.47 \ R$.
Ignition wake	FIXED (firmware)	The $\div 34$ divider cannot produce a logic edge, so IGN cannot wake the MCU by EXTI: RTC-timed ADC polling (FW-14). Hardware alternative for a later revision: a comparator or a $100 \text{ k} / 9.1 \text{ k}$ divider with a 5.1 V zener.
Digital inputs are not galvanically isolated	OK (documented)	The opto is used as a 100 V level shifter; the input return is the unit's GND. Handoff and installation sheet corrected ("optically isolated" removed).
SIM ESD array capacitance	VERIFY	PESD5V0L5UY $\leq 19 \text{ pF}$ vs. Quectel's 15 pF guideline; nearest in-stock SOT-363 part. Check SIM communication on the first units; a 4-line 0.5 pF array on a different footprint is the fallback.
DRC warnings (11)	OK	8 silk marks on the DNP eSIM pads, 1 silk text near D2, 2 KiCad "dangling" flags on connected T-junctions. No effect on manufacture.
ERC warnings (23)	OK	Rails carrying both a local and a hierarchical label (3V3, 5V0, SYS, VIN, VIN_P ...) and single-pin sense labels; the netlist confirms every rail is one net.
Bench and certification items in sections 6 and 7	NOT OK	Owner: hardware bring-up (bench), certification lab (EMC, radio, IP, safety), pack vendor (battery certificates).

9. Deliverables

- Manufacturing package (one zip, ready for JLCPCB or a local fab): `out/telematics-tracker-mfg-2026-09-30-<commit>.zip` — 1_PCB_fabrication (Gerbers, drill, ODB++, IPC-2581, fabrication drawing), 2_PCB_assembly (BOM with LCSC and manufacturer part numbers, pick-and-place, assembly drawings), 3_documentation (schematic PDF, assembled-board STEP, BOM audit, this report), 4_enclosure (STEP/STL/Blender for both housing variants), README.
- Sources: `tools/sheets.py` (schematic generator), `telematics-tracker.kicad_pcb`, `git tag v1.0.2-verified-2026-09-30` on GitHub (Immanueldavidckp/schematics).
- Documents corrected in this review: `docs/firmware-notes.md` (FW-6/8/13/14/17, FW-18...22, BV-1, BV-5...7), `docs/telematics-handoff.md`, `docs/installation-sheet.md`, `design-log.md` (entry 2026-09-30).

Method note: every finding was confirmed against the manufacturer document, not the KiCad symbol; where the two disagreed (BAV99, D_TVS) the datasheet won. All board changes were applied through DRC-gated scripts (`tools/manual_routes.py`, `tools/board_fixes.py`) so that no change could leave the board with more violations than before.